Second: Homework – Model Answers P. 19

(1) Answer the following:

a. Explain the function of the muscles lining the inner wall of the small intestine.

To move food along the intestine and mix it with digestive juices, helping digestion and absorption.

b. Compare the functions of the small intestine and the large intestine in detail.

Small intestine: completes most digestion and absorbs most nutrients into the blood.

Large intestine: absorbs water from undigested food and forms faeces.

c. What would happen if the muscles of the small intestine were removed? What effect would this have on the body?

Food would not move and mix properly through the small intestine. Digestion and nutrient absorption would be reduced, causing serious problems for the body.

d. How does the circulatory system support the digestive system?

Blood supplies the digestive organs with oxygen and carries absorbed nutrients from the small intestine to cells throughout the body.

(2) Complete the following:

a. Most of the products of digestion are absorbed in the small intestine.

b. Villi help increase the surface area for absorption in the small intestine.

(3) Explain the function of the heart in the body. Why is the heart considered one of the most important organs of the circulatory system?

The heart is a muscular pump that circulates blood. It sends oxygen and nutrients to body cells and carries carbon dioxide and other wastes away.

(4) Give reason for each of the following:

a. The lining of the small intestine contains villi.

Villi increase the surface area, allowing digested nutrients to be absorbed efficiently.

b. A large network of blood capillaries surrounds the wall of the small intestine.

The capillaries quickly carry absorbed nutrients away to the rest of the body.

(5) Why is the liver one of the most important organs in the body?

The liver produces bile to help digest fats, processes and stores nutrients, and breaks down harmful substances in the blood.